

Lagrange Duality

Primal problem:

$$\min_x f_0(x)$$

$$f_i(x) \leq 0, \quad i = 1, \dots, n$$

$$h_j(x) = 0, \quad j = 1, \dots, m$$

$$\lambda_i$$

$$\nu_j$$

Lagrangian:

$$L(x, \lambda, \nu) := f_0(x) + \sum_{i=1}^n \lambda_i f_i(x) + \sum_{j=1}^m \nu_j h_j(x)$$

Dual objective:

$$g(\lambda, \nu) := \inf_x L(x, \lambda, \nu)$$

Dual problem:

$$\max_{\lambda, \nu} g(\lambda, \nu)$$

$$g(\lambda, \nu) > -\infty$$

$$\lambda \geq 0$$

Build the Dual Problem

Build the dual, solve primal and dual, compare their optimal values:

$$\min x_1$$

$$x_1 - x_2 = 1$$

$$x_2 \geq 2$$

Complementary Slackness

Let $-\infty < p^* = f_0(x^*) = d^* = g(\lambda^*, \nu^*) < \infty$

Complementary Slackness

Let $-\infty < p^* = f_0(x^*) = d^* = g(\lambda^*, \nu^*) < \infty$

$$\begin{aligned} f_0(x^*) = g(\lambda^*, \nu^*) &= \inf_x L(x^*, \lambda^*, \nu^*) \\ &\leq f_0(x^*) + \sum_{i=1}^n \lambda_i^* f_i(x^*) + \sum_{j=1}^m \underbrace{\nu_j^* h_j(x^*)}_{=0} \leq f_0(x^*) \end{aligned}$$

Complementary Slackness

$$\text{Let} \quad -\infty < p^* = f_0(x^*) = d^* = g(\lambda^*, \nu^*) < \infty$$

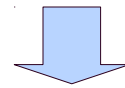
$$\begin{aligned} f_0(x^*) = g(\lambda^*, \nu^*) &= \inf_x L(x^*, \lambda^*, \nu^*) \\ &\leq f_0(x^*) + \sum_{i=1}^n \lambda_i^* f_i(x^*) + \sum_{j=1}^m \underbrace{\nu_j^* h_j(x^*)}_{=0} \leq f_0(x^*) \end{aligned}$$

$$\sum_{i=1}^n \underbrace{\lambda_i^*}_{\geq 0} \underbrace{f_i(x^*)}_{\leq 0} = 0 \quad \Rightarrow \quad \lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m$$

$$f_i(x^*) = 0 \quad \text{or} \quad \lambda_i^* = 0$$

Karush-Kuhn-Tucker (KKT) conditions

Let $-\infty < p^* = f_0(x^*) = d^* = g(\lambda^*, \nu^*) < \infty$

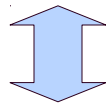


(necessary optimality conditions)

$$\left. \begin{array}{l} f_i(x^*) \leq 0, \quad i = 1, \dots, n \\ h_j(x^*) = 0, \quad j = 1, \dots, m \end{array} \right\} \text{Primal feasibility}$$

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, n \quad \left. \vphantom{\lambda_i^* f_i(x^*) = 0} \right\} \text{Complementary slackness}$$

$$\left. \begin{array}{l} \lambda_i^* \geq 0, \quad i = 1, \dots, n \\ \nabla f_0(x^*) + \sum_{i=1}^n \lambda_i^* \nabla f_i(x^*) \\ + \sum_{j=1}^m \nu_j^* \nabla h_j(x^*) = 0 \end{array} \right\} \text{Dual feasibility and optimality}$$



$$\inf_x L(x, \lambda, \nu)$$

Subgradients are used in place of gradients in non-differentiable case.

Karush-Kuhn-Tucker (KKT) conditions

$$\left. \begin{array}{l} f_i(x^*) \leq 0, \quad i = 1, \dots, n \\ h_j(x^*) = 0, \quad j = 1, \dots, m \end{array} \right\} \text{Primal feasibility}$$

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, n \quad \left. \right\} \text{Complementary slackness}$$

$$\left. \begin{array}{l} \lambda_i^* \geq 0, \quad i = 1, \dots, n \\ \nabla f_0(x^*) + \sum_{i=1}^n \lambda_i^* \nabla f_i(x^*) \\ + \sum_{j=1}^m \nu_j^* \nabla h_j(x^*) = 0 \end{array} \right\} \text{Dual feasibility and optimality}$$

Theorem: Let f_i — convex differentiable
 h_j — linear

Then KKT conditions are necessary and sufficient for optimality

Karush-Kuhn-Tucker (KKT) conditions

$$\left. \begin{array}{l} f_i(x^*) \leq 0, \quad i = 1, \dots, n \\ h_j(x^*) = 0, \quad j = 1, \dots, m \end{array} \right\} \text{Primal feasibility}$$

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, n \quad \left. \right\} \text{Complementary slackness}$$

$$\left. \begin{array}{l} \lambda_i^* \geq 0, \quad i = 1, \dots, n \\ \nabla f_0(x^*) + \sum_{i=1}^n \lambda_i^* \nabla f_i(x^*) \\ + \sum_{j=1}^m \nu_j^* \nabla h_j(x^*) = 0 \end{array} \right\} \text{Dual feasibility and optimality}$$

Theorem: Let f_i — convex differentiable
 h_j — linear

Then KKT conditions are sufficient for optimality

Proof:

$$g(\lambda^*, \nu^*) = L(x^*, \lambda^*, \nu^*) = f_0(x^*) + \sum_{i=1}^n \lambda_i^* f_i(x^*) + \sum_{j=1}^m \nu_j^* h_j(x^*) = f(x^*)$$

KKT for LP: Necessary and Sufficient for Optimality

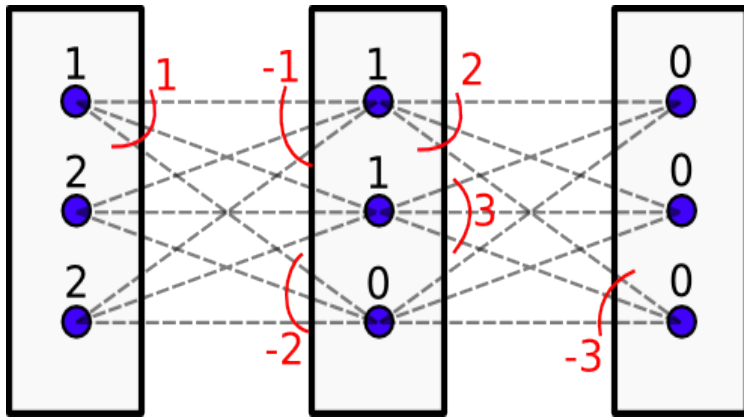
$$\left. \begin{array}{l} f_i(x^*) \leq 0, \quad i = 1, \dots, n \\ h_j(x^*) = 0, \quad j = 1, \dots, m \end{array} \right\} \text{Primal feasibility}$$
$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, n \quad \left. \vphantom{\lambda_i^* f_i(x^*) = 0} \right\} \text{Complementary slackness}$$
$$\left. \begin{array}{l} \lambda_i^* \geq 0, \quad i = 1, \dots, n \\ g(\lambda, \nu) > -\infty \end{array} \right\} \text{Dual feasibility}$$

Corollary: For linear problems any feasible primal/dual pair satisfying complementary slackness constraints is optimal

Proof:

$$\nabla_x L(x, \lambda, \nu) = \nabla_x \left(-\langle \nu, b \rangle + \langle c - \lambda + A^\top \nu, x \rangle \right) = c - \lambda + A^\top \nu = 0$$

Recall: Reparametrization

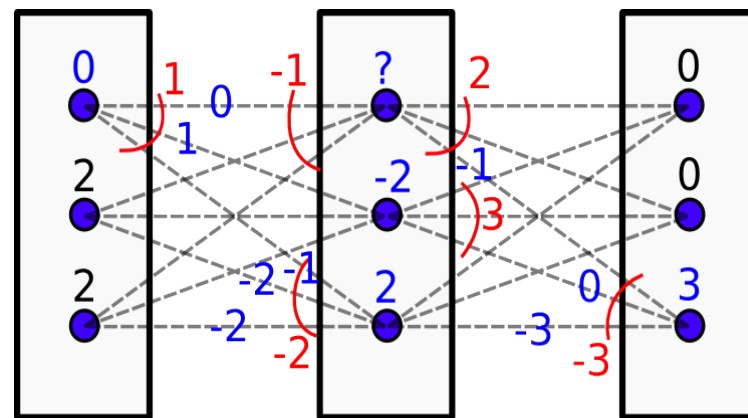
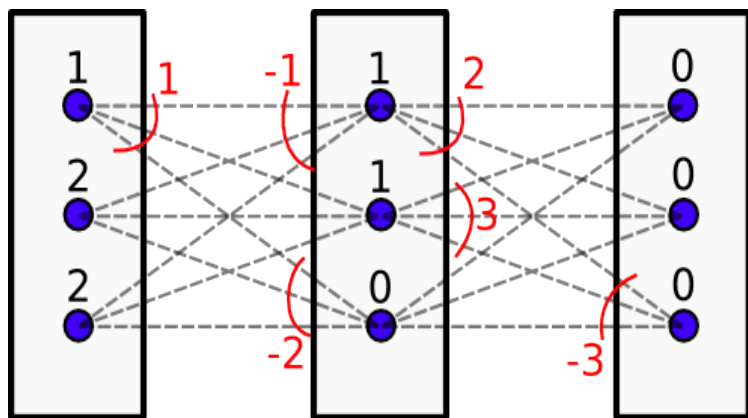


$$\sum_{v \in \mathcal{V}} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v) = \sum_{v \in \mathcal{V}} \theta_v^\phi(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}^\phi(x_u, x_v)$$

$$\theta_v^\phi(x_v) := \theta_v(x_v) - \sum_{u \in \mathcal{N}_b(v)} \phi_{v,u}(x_v)$$

$$\theta_{uv}^\phi(x_u, x_v) := \theta_{uv}(x_u, x_v) + \phi_{v,u}(x_v) + \phi_{u,v}(x_u)$$

Recall: Reparametrization

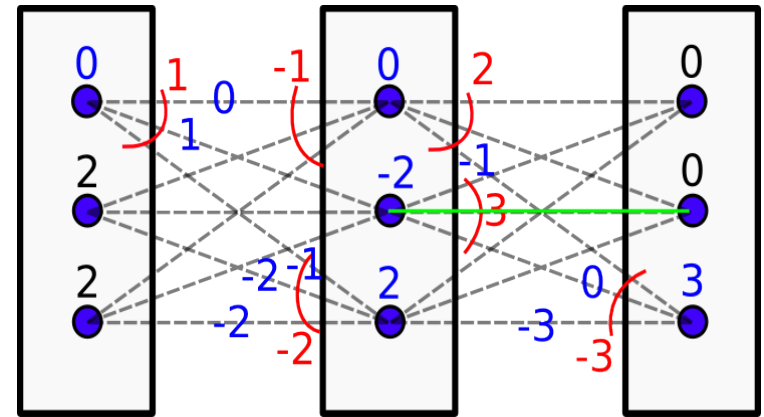
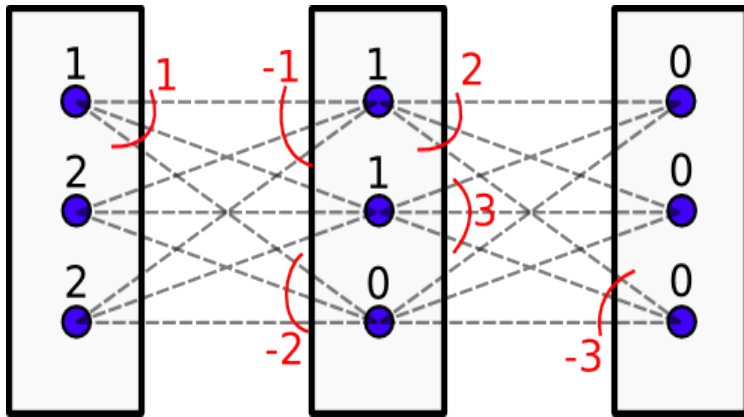


$$\sum_{v \in \mathcal{V}} \theta_v(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}(x_u, x_v) = \sum_{v \in \mathcal{V}} \theta_v^\phi(x_v) + \sum_{uv \in \mathcal{E}} \theta_{uv}^\phi(x_u, x_v)$$

$$\theta_v^\phi(x_v) := \theta_v(x_v) - \sum_{u \in \mathcal{N}_b(v)} \phi_{v,u}(x_v)$$

$$\theta_{uv}^\phi(x_u, x_v) := \theta_{uv}(x_u, x_v) + \phi_{v,u}(x_v) + \phi_{u,v}(x_u)$$

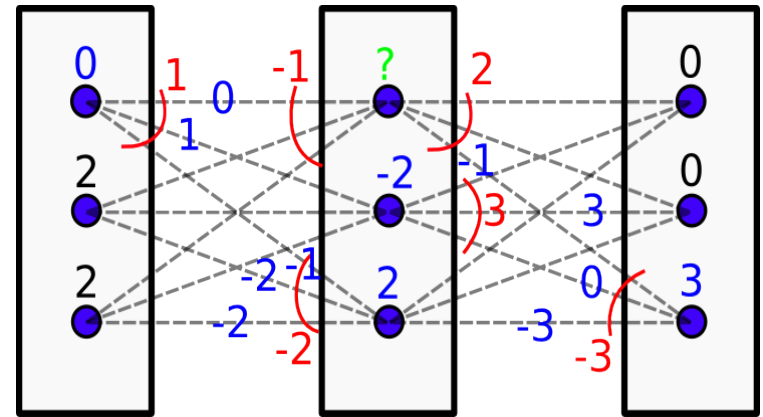
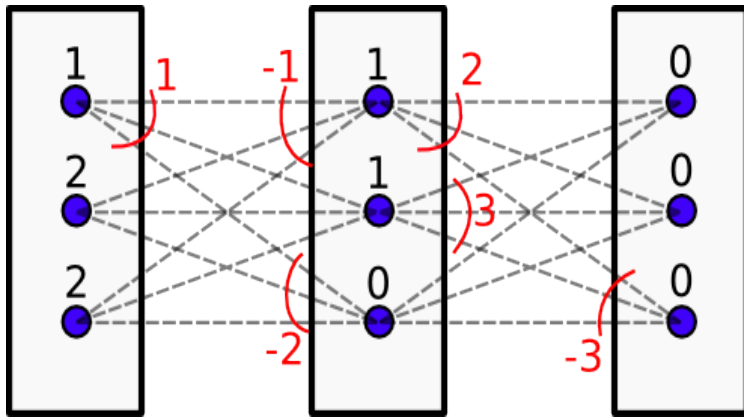
Recall: Reparametrization



What is the reparametrized potential on the green edge?

- 1) 0
- 2) 4
- 3) -1
- 4) 3
- 5) 1
- 6) None is correct

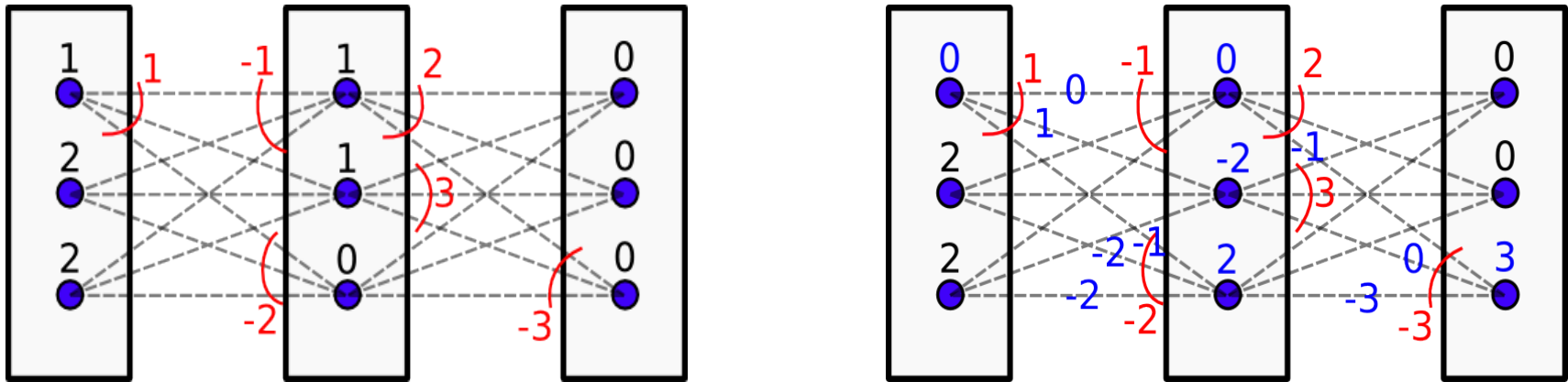
Recall: Reparametrization



What is the reparametrized potential on the label with a question mark?

- 1) 0
- 2) 4
- 3) -1
- 4) 3
- 5) 1
- 6) None is correct

Recall: Reparametrization



$$D(\phi) := \sum_{v \in \mathcal{V}} \min_{s \in \mathcal{X}_v} \theta_v^\phi(x_v) + \sum_{uv \in \mathcal{E}} \min_{(s,t) \in \mathcal{X}_{uv}} \theta_{uv}^\phi(x_u, x_v)$$

What is the current dual value ?

- 1) 0
- 2) 11
- 3) -7
- 4) -4
- 5) -5
- 6) None is correct

Complementary Slackness

$$\min_{\mu} \langle \theta, \mu \rangle$$

$$\begin{array}{ll} & 1 - \sum_{x_v \in \mathcal{X}_v} \mu_v(x_v) = 0, \quad v \in \mathcal{V} \\ & 1 - \sum_{(x_u, x_v) \in \mathcal{X}_{uv}} \mu_{uv}(x_u, x_v) = 0, \quad uv \in \mathcal{E} \\ \text{s.t.} & \sum_{x_v \in \mathcal{X}_v} \mu_{uv}(x_u, x_v) - \mu_u(x_u) = 0, \quad x_u \in \mathcal{X}_u, \quad uv \in \mathcal{E} \\ & \sum_{x_u \in \mathcal{X}_u} \mu_{uv}(x_u, x_v) - \mu_v(x_v) = 0, \quad x_v \in \mathcal{X}_v, \quad uv \in \mathcal{E} \\ & -\mu \leq 0 \end{array} \quad \left| \begin{array}{l} \gamma_v \\ \gamma_{uv} \\ \phi_{v,u}(x_v) \\ \phi_{u,v}(x_u) \\ \beta \in \mathbb{R}^I \end{array} \right.$$

$$\max_{\gamma, \phi, \beta} \sum_{v \in \mathcal{V}} \gamma_v + \sum_{uv \in \mathcal{E}} \gamma_{uv}$$

$$\gamma_v = \theta_v^\phi(x_v) - \beta_v(x_v), \quad v \in \mathcal{V}, \quad x_v \in \mathcal{X}_v$$

$$\gamma_{uv} = \theta_{uv}^\phi(x_u, x_v) - \beta_{uv}(x_u, x_v), \quad uv \in \mathcal{E}, \quad (x_u, x_v) \in \mathcal{X}_{uv}$$

$$\beta \geq 0$$

Complementary Slackness

$$\begin{aligned}
 & \min_{\mu} \langle \theta, \mu \rangle \\
 & \text{s.t.} \quad \begin{array}{l|l}
 1 - \sum_{x_v \in \mathcal{X}_v} \mu_v(x_v) = 0, & v \in \mathcal{V} & \gamma_v \\
 1 - \sum_{(x_u, x_v) \in \mathcal{X}_{uv}} \mu_{uv}(x_u, x_v) = 0, & uv \in \mathcal{E} & \gamma_{uv} \\
 \sum_{x_v \in \mathcal{X}_v} \mu_{uv}(x_u, x_v) - \mu_u(x_u) = 0, & x_u \in \mathcal{X}_u, uv \in \mathcal{E} & \phi_{v,u}(x_v) \\
 \sum_{x_u \in \mathcal{X}_u} \mu_{uv}(x_u, x_v) - \mu_v(x_v) = 0, & x_v \in \mathcal{X}_v, uv \in \mathcal{E} & \phi_{u,v}(x_u) \\
 -\mu \leq 0 & & \beta \in \mathbb{R}^I
 \end{array} \\
 & \max_{\gamma, \phi, \beta} \sum_{v \in \mathcal{V}} \gamma_v + \sum_{uv \in \mathcal{E}} \gamma_{uv} \\
 & \gamma_v = \theta_v^\phi(x_v) - \beta_v(x_v), \quad v \in \mathcal{V}, \quad x_v \in \mathcal{X}_v \\
 & \gamma_{uv} = \theta_{uv}^\phi(x_u, x_v) - \beta_{uv}(x_u, x_v), \quad uv \in \mathcal{E}, \quad (x_u, x_v) \in \mathcal{X}_{uv} \\
 & \beta \geq 0 \\
 & \beta_w(s) \mu_w(s) = 0
 \end{aligned}$$

Complementary Slackness

$$\max_{\gamma, \phi, \beta} \sum_{v \in \mathcal{V}} \gamma_v + \sum_{uv \in \mathcal{E}} \gamma_{uv}$$

$$\gamma_v = \theta_v^\phi(x_v) - \beta_v(x_v), \quad v \in \mathcal{V}, \quad x_v \in \mathcal{X}_v$$

$$\gamma_{uv} = \theta_{uv}^\phi(x_u, x_v) - \beta_{uv}(x_u, x_v), \quad uv \in \mathcal{E}, \quad (x_u, x_v) \in \mathcal{X}_{uv}$$

$$\beta \geq 0$$

$$\beta_w(s) \mu_w(s) = 0$$

$$(\beta_v(x_v) = 0) \quad \Rightarrow \quad \gamma_v = \theta_v^\phi(x_v) = \min_{s \in \mathcal{X}_v} \theta_v^\phi(s)$$

Complementary Slackness

$$\max_{\gamma, \phi, \beta} \sum_{v \in \mathcal{V}} \gamma_v + \sum_{uv \in \mathcal{E}} \gamma_{uv}$$

$$\gamma_v = \theta_v^\phi(x_v) - \beta_v(x_v), \quad v \in \mathcal{V}, \quad x_v \in \mathcal{X}_v$$

$$\gamma_{uv} = \theta_{uv}^\phi(x_u, x_v) - \beta_{uv}(x_u, x_v), \quad uv \in \mathcal{E}, \quad (x_u, x_v) \in \mathcal{X}_{uv}$$

$$\beta \geq 0$$

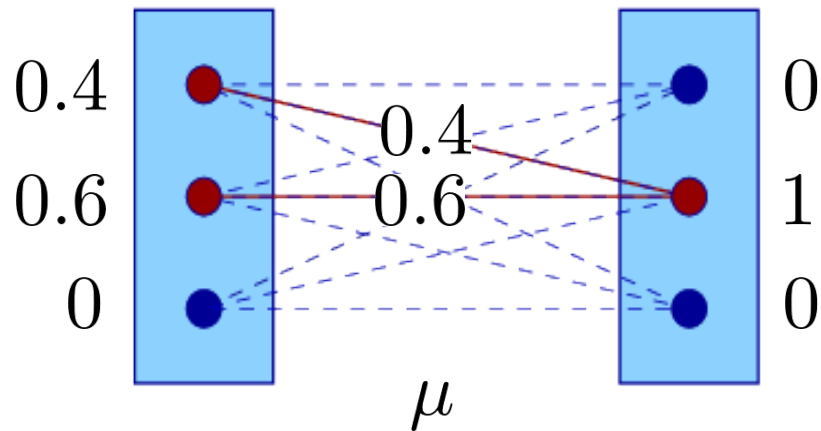
$$\beta_w(s) \mu_w(s) = 0$$

$$(\beta_v(x_v) = 0) \quad \Rightarrow \quad \gamma_v = \theta_v^\phi(x_v) = \min_{s \in \mathcal{X}_v} \theta_v^\phi(s)$$

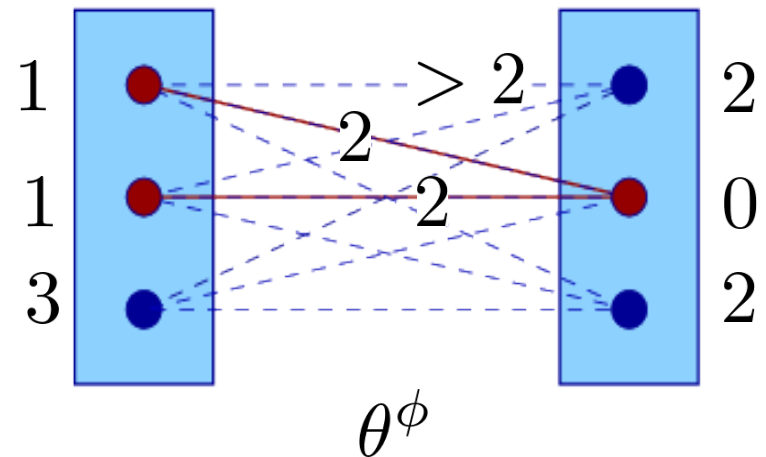
$$(\beta_{uv}(x_u, x_v) = 0) \quad \Rightarrow \quad \gamma_{uv} = \theta_{uv}^\phi(x_u, x_v) = \min_{st \in \mathcal{X}_{uv}} \theta_{uv}^\phi(s, t)$$

Complementary Slackness

Primal:



Dual:



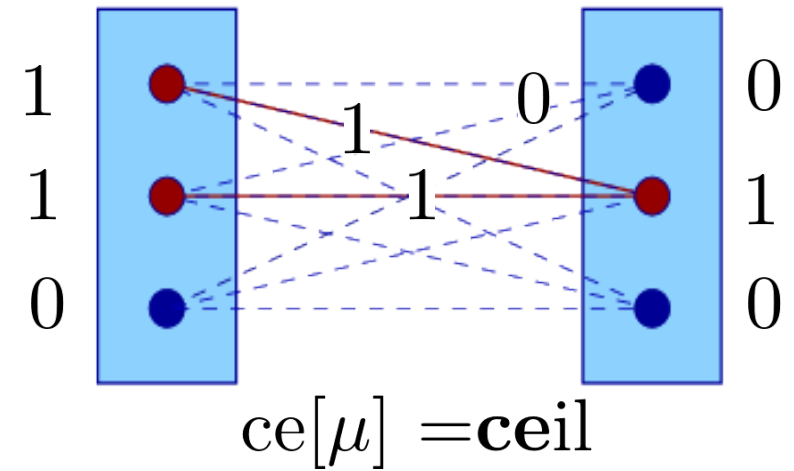
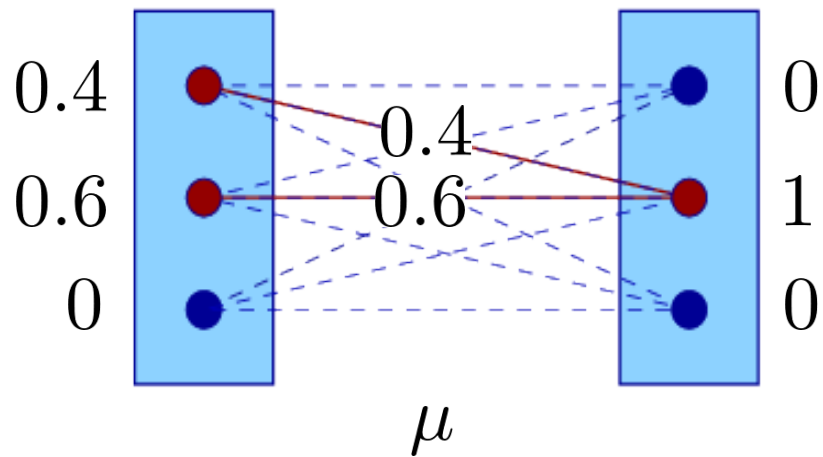
$$\beta_w(s)\mu_w(s) = 0$$

$$(\mu_v(x_v) > 0) \Rightarrow \theta_v^\phi(x_v) = \min_{s \in \mathcal{X}_v} \theta_v^\phi(s)$$

$$(\mu_{uv}(x_u, x_v) > 0) \Rightarrow \theta_v^\phi(x_u, x_v) = \min_{st \in \mathcal{X}_{uv}} \theta_v^\phi(s, t)$$

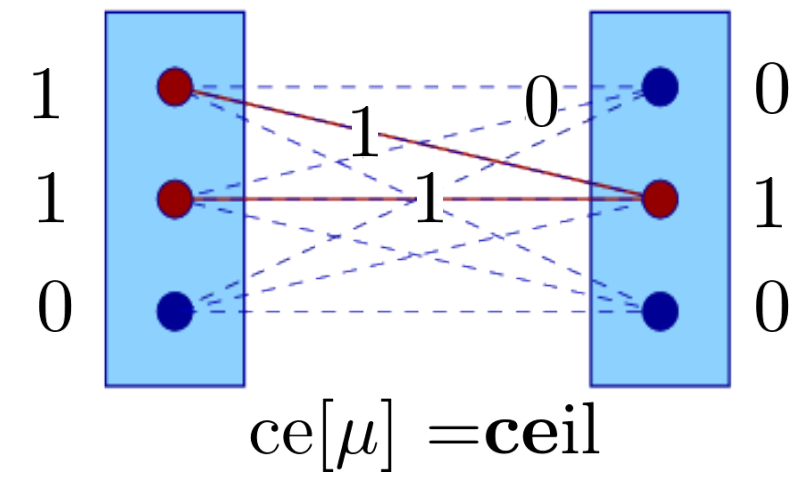
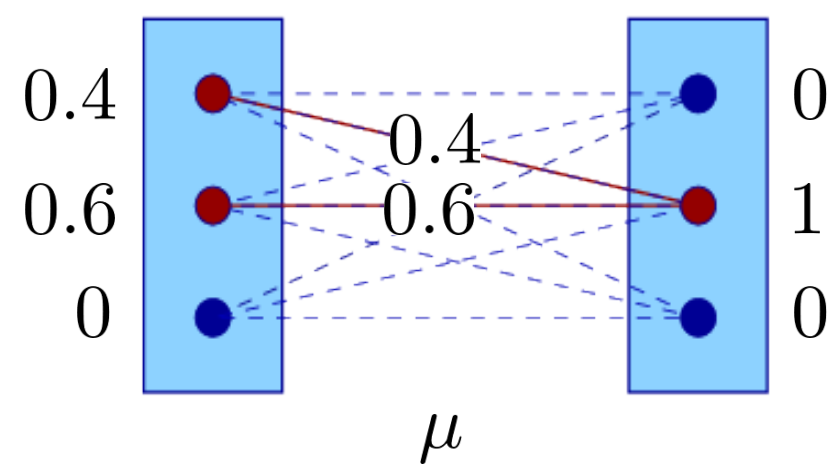
Binarization of Primal

Primal:

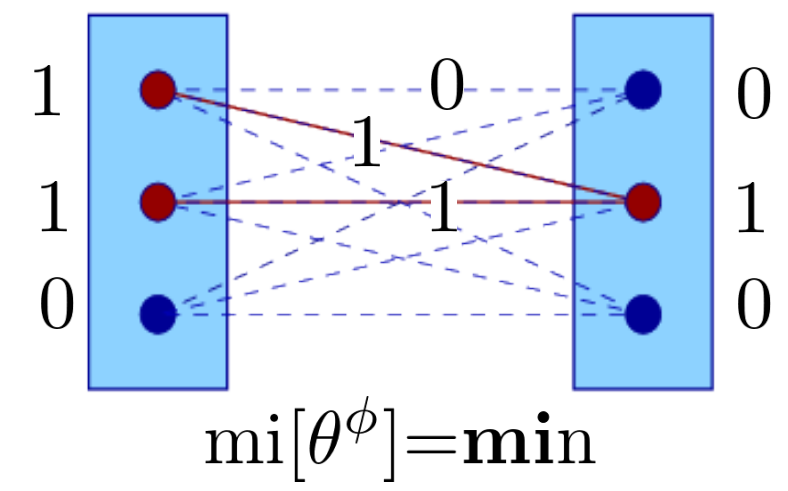
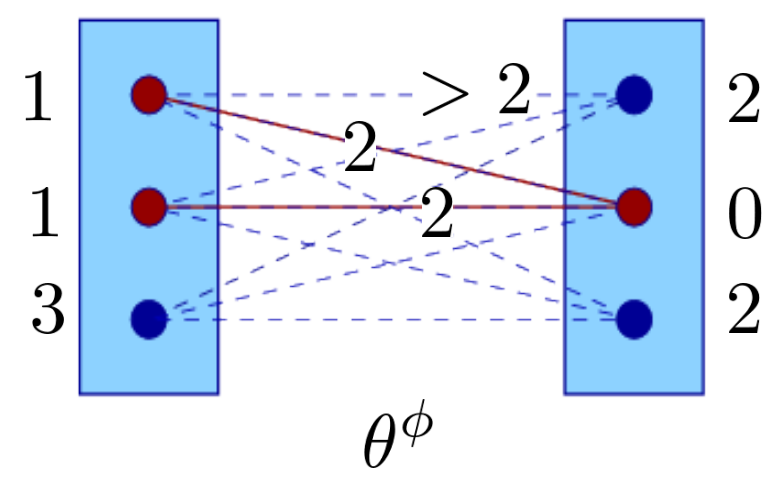


Binarization of Primal and Dual Vectors

Primal:

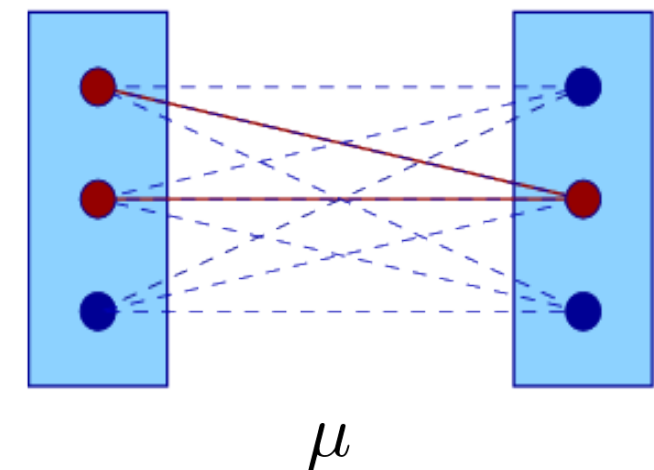
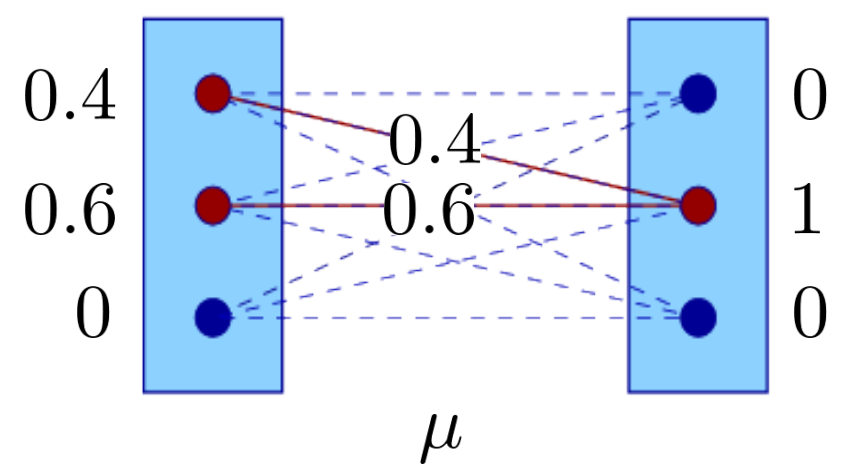


Dual:

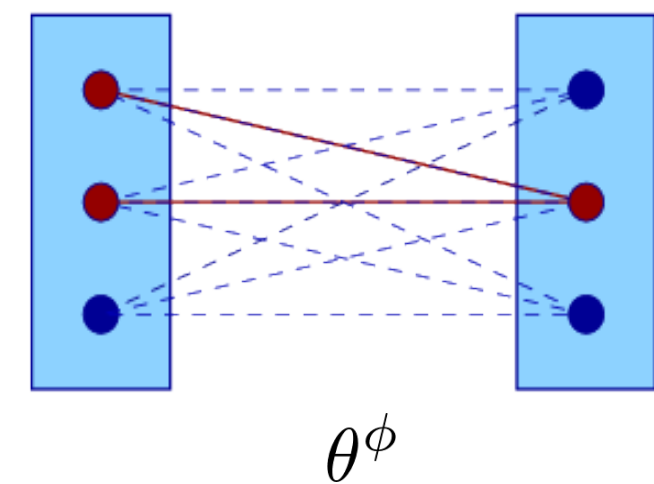
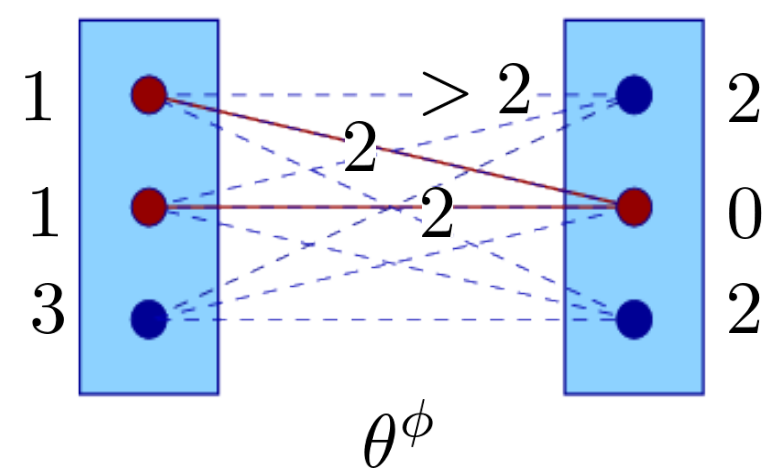


Binarization of Primal and Dual Vectors: Notation

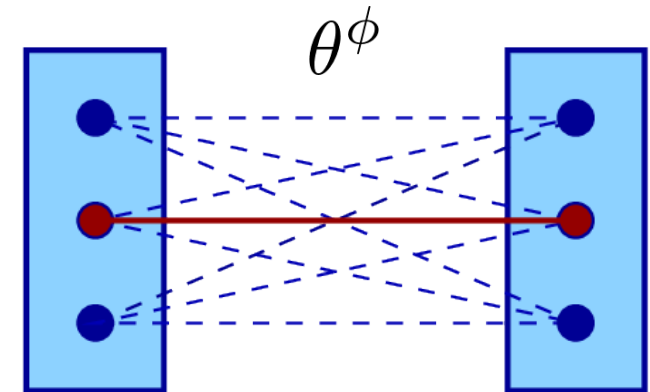
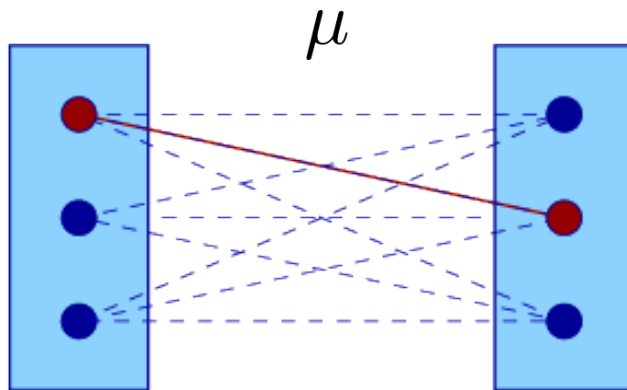
Primal:



Dual:



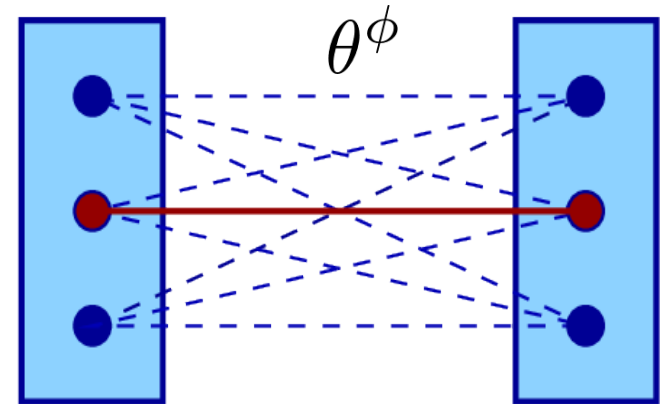
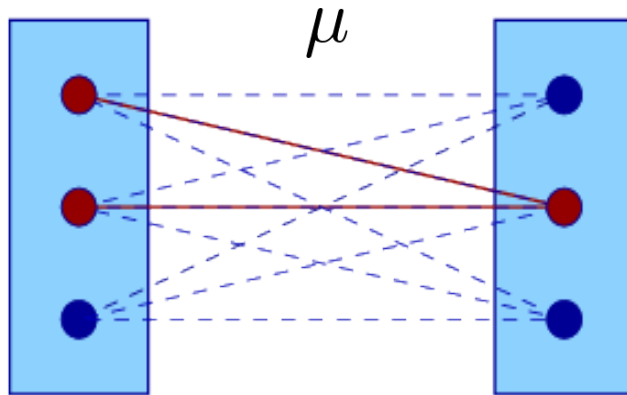
Complementary Slackness: Example 1



Can this primal-dual pair be optimal?

- 1) Yes, because the complementary slackness condition is satisfied
- 2) Yes, the complementary slackness condition is not satisfied, but it is not necessary for optimum
- 3) Yes, because μ is integer and $\theta_v^\phi, \theta_u^\phi$ have unique minima
- 4) No, because the complementary slackness condition is not satisfied
- 5) No, because optimal μ can not be integer, since it corresponds to the relaxed problem
- 6) None is correct

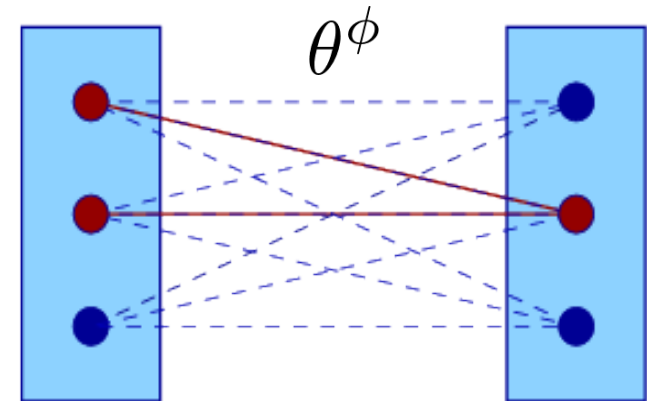
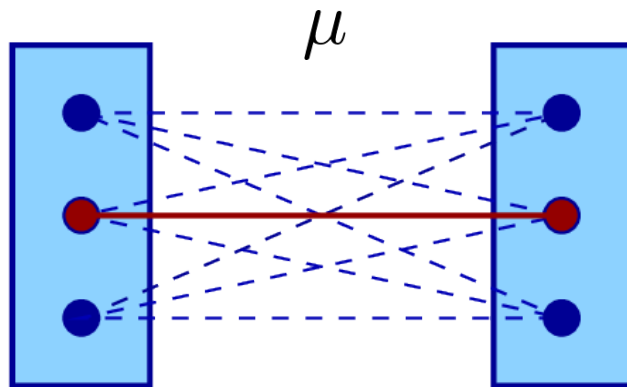
Complementary Slackness: Example 2



Can this primal-dual pair be optimal?

- 1) Yes, because the complementary slackness condition is satisfied
- 2) Yes, the complementary slackness condition is not satisfied, but it is not necessary for optimum
- 3) Yes, because μ is integer and $\theta_v^\phi, \theta_u^\phi$ have unique minima
- 4) No, because the complementary slackness condition is not satisfied
- 5) No, because optimal μ must be integer
- 6) None is correct

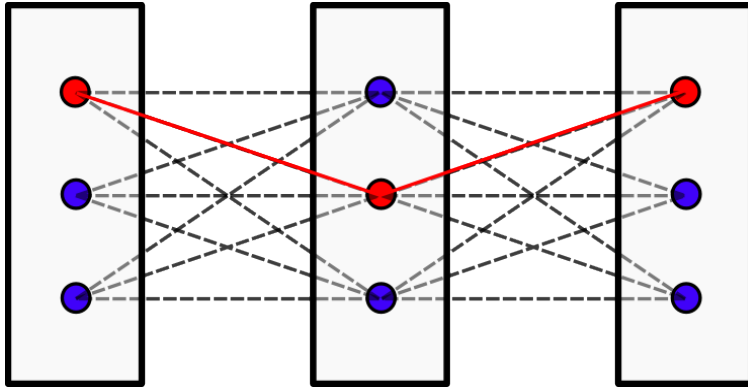
Complementary Slackness: Example 3



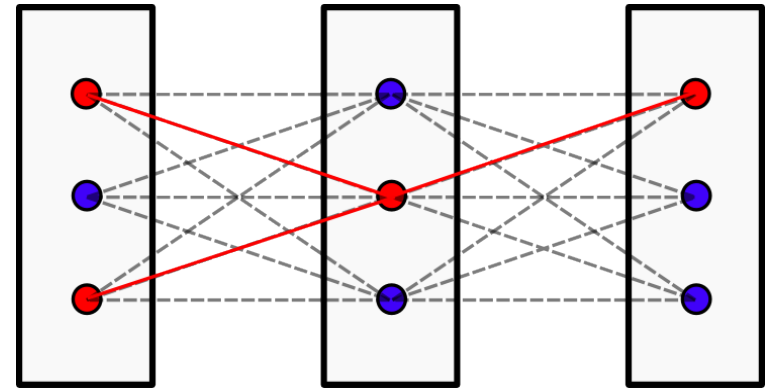
Can this primal-dual pair be optimal?

- 1) Yes, because the complementary slackness condition is satisfied
- 2) Yes, the complementary slackness condition is not satisfied, but it is not necessary for optimum
- 3) Yes, because μ is integer and $\theta_v^\phi, \theta_u^\phi$ have unique minima
- 4) No, because the complementary slackness condition is not satisfied
- 5) No, because optimal $\theta_v^\phi, \theta_u^\phi$ must have unique minima
- 6) None is correct

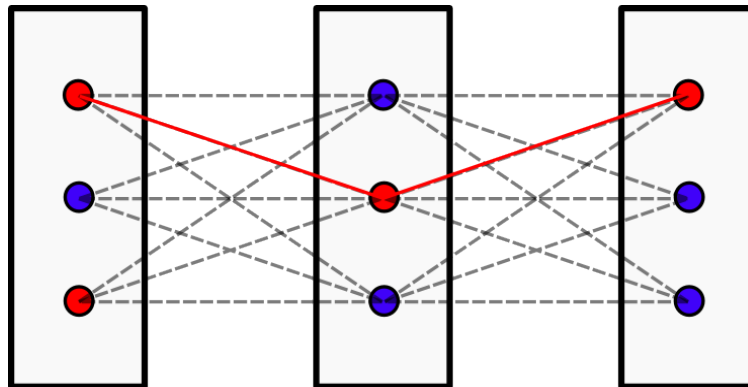
Arc Consistency



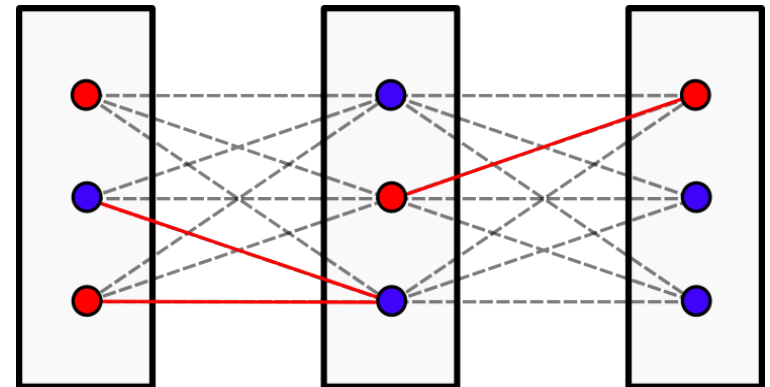
strict arc-consistency



arc-consistency



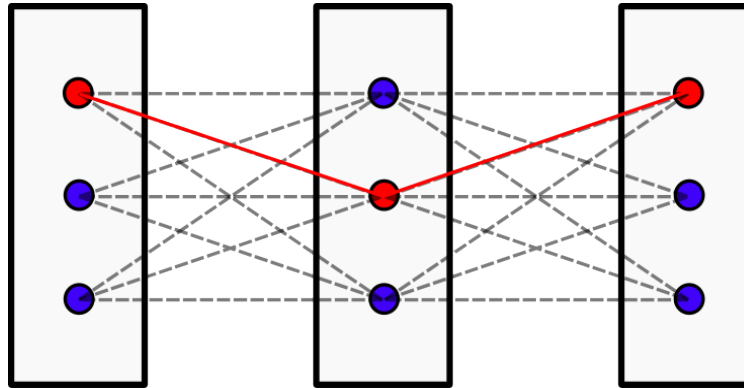
no arc-consistency



no arc-consistency

Arc Consistency

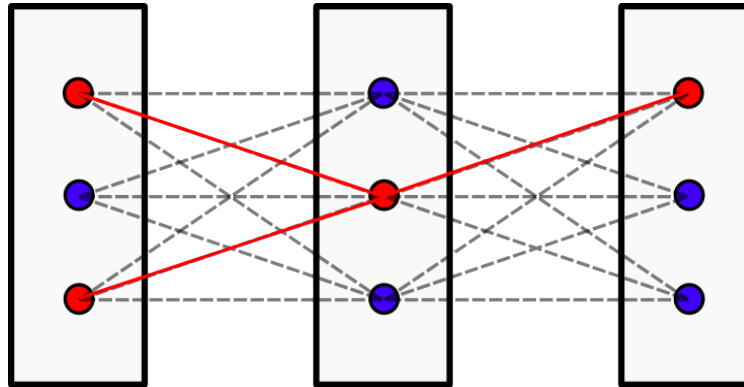
Is any $\delta(x)$, $x \in \mathcal{X}_V$ arc-consistent ?



- 1) Yes
- 2) No
- 3) No idea

Arc Consistency

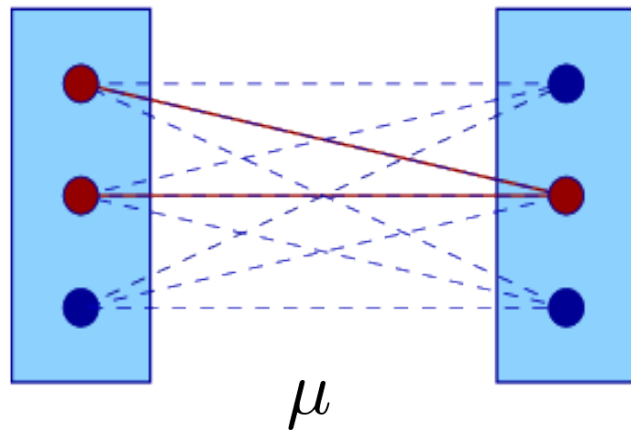
Is any $ce[\mu]$, $\mu \in \mathcal{L}$ arc-consistent ?



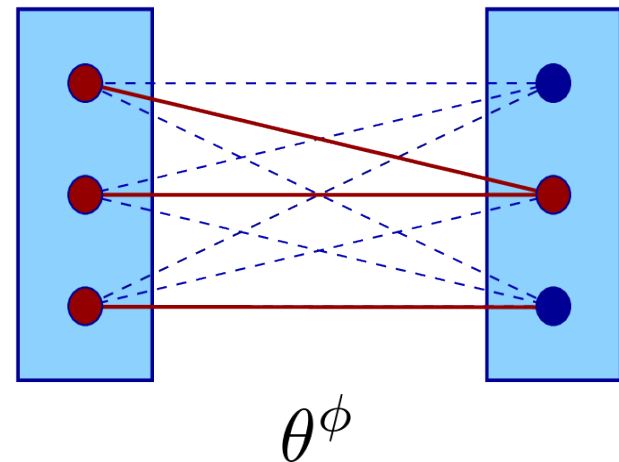
- 1) Yes
- 2) No
- 3) No idea

Recall: Complementary Slackness

Primal:



Dual:



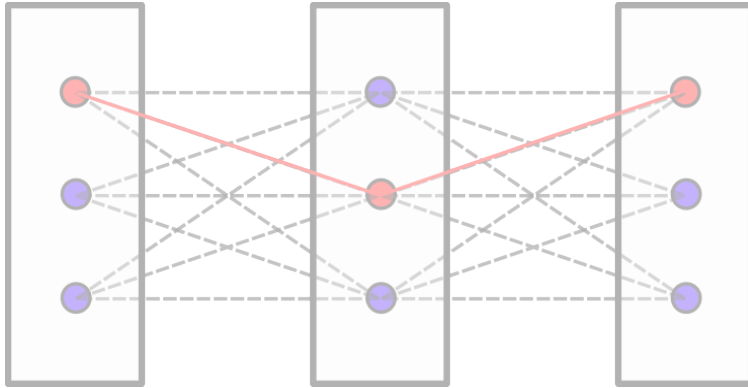
$$\beta_w(s)\mu_w(s) = 0$$

$$(\mu_v(x_v) > 0) \Rightarrow \theta_v^\phi(x_v) = \min_{s \in \mathcal{X}_v} \theta_v^\phi(s)$$

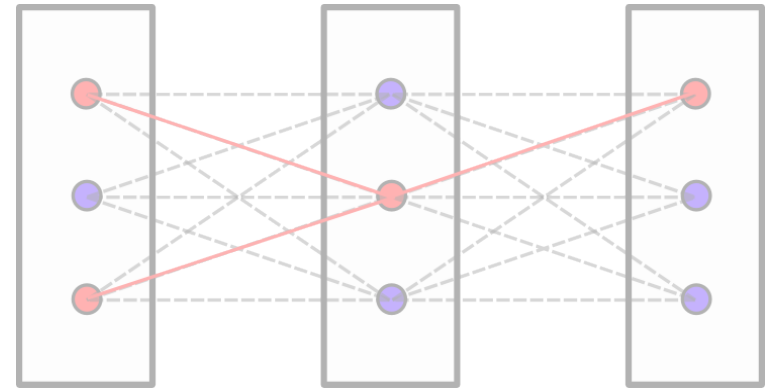
$$(\mu_{uv}(x_u, x_v) > 0) \Rightarrow \theta_v^\phi(x_u, x_v) = \min_{st \in \mathcal{X}_{uv}} \theta_v^\phi(s, t)$$

Dual binarized vector must contain an arc-consistent subset – kernel!

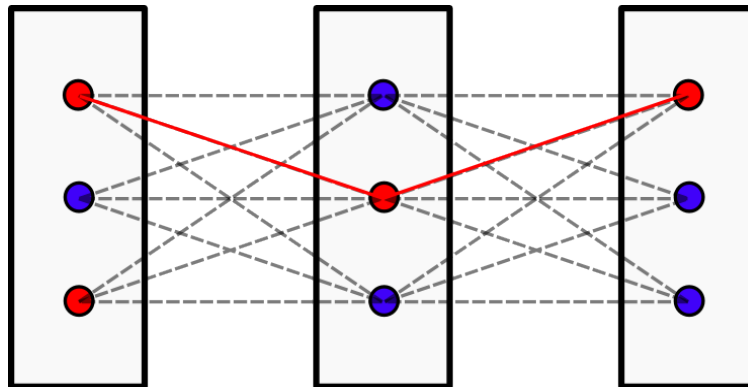
Arc Consistency



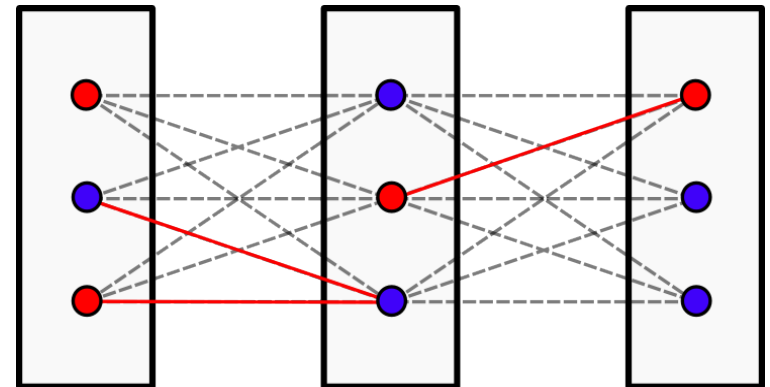
strict arc-consistency



arc-consistency

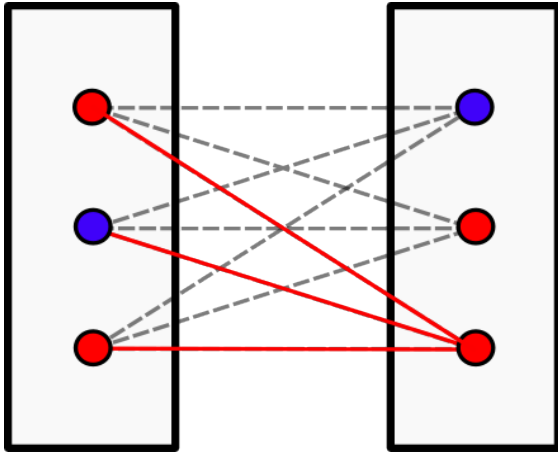


non-empty kernel



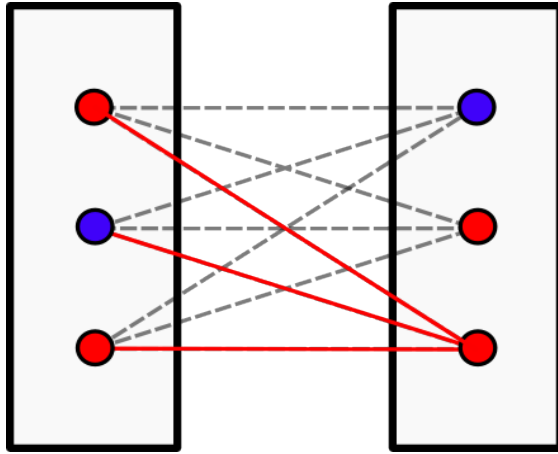
empty kernel

Kernel as the Maximal Arc-Consistent Subset



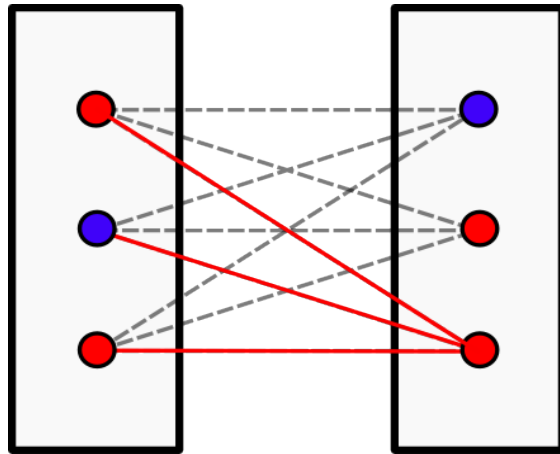
- 1) arc-consistent
- 2) not arc-consistent
- 3) no idea

Kernel as the Maximal Arc-Consistent Subset

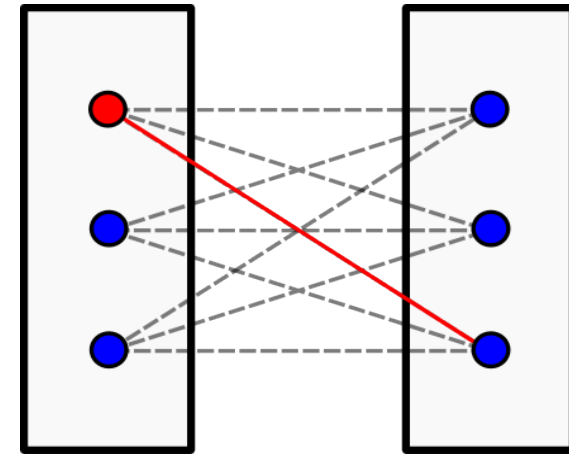
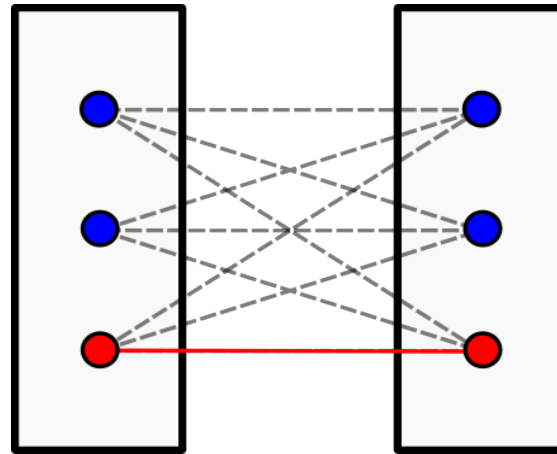


not arc-consistent

Kernel as the Maximal Arc-Consistent Subset

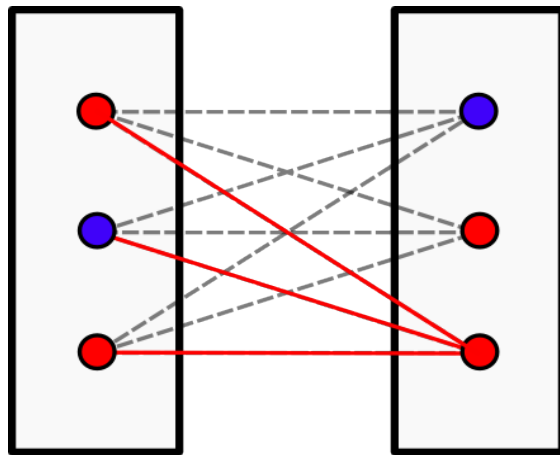


not arc-consistent

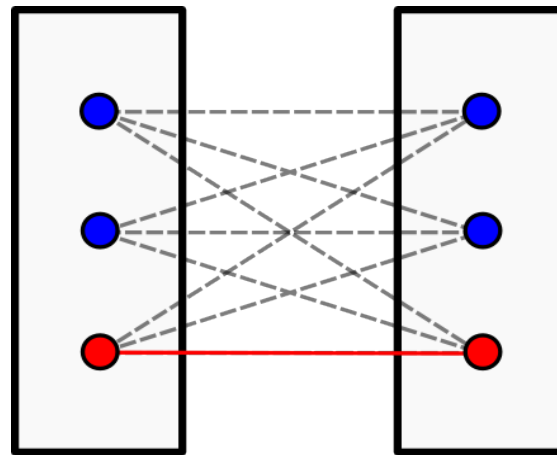


- 1) left a-c, right not-a-c
- 2) right a-c, left – no-a-c
- 3) both not-a-c
- 4) both strict-a-c
- 5) both a-c
- 6) no idea

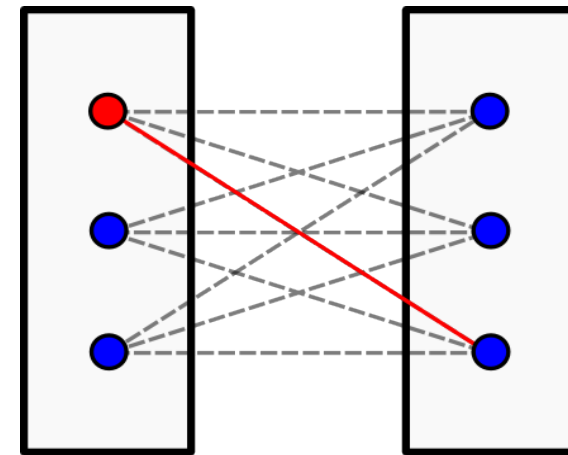
Kernel as the Maximal Arc-Consistent Subset



inconsistent

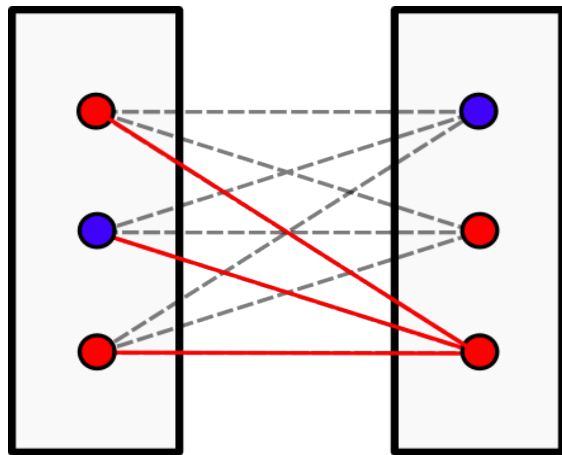


Arc-consistent subset

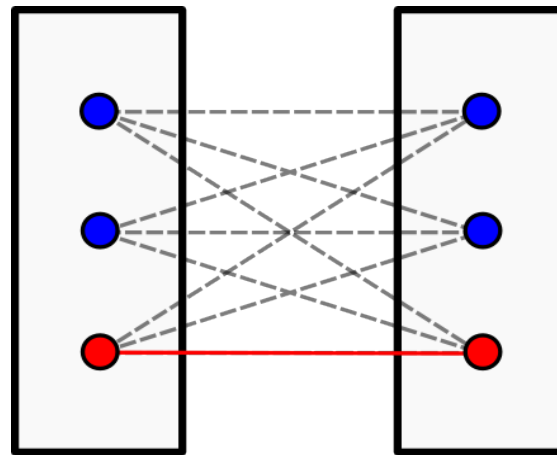


Arc-consistent subset

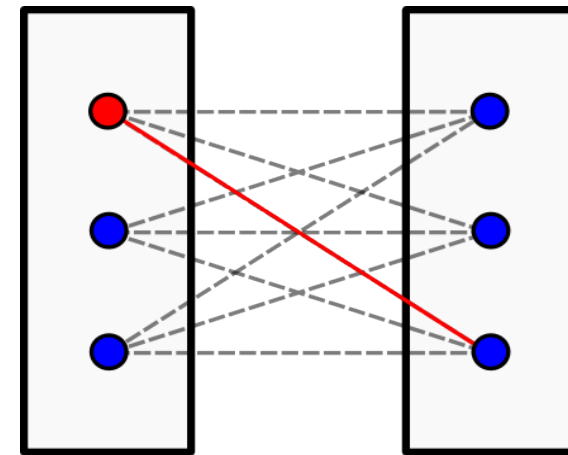
Kernel as the Maximal Arc-Consistent Subset



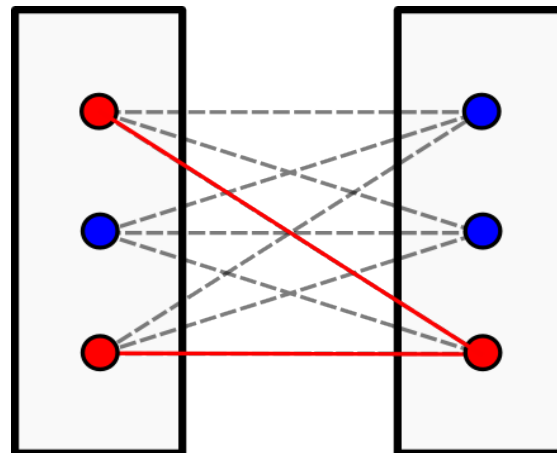
inconsistent



Arc-consistent subset



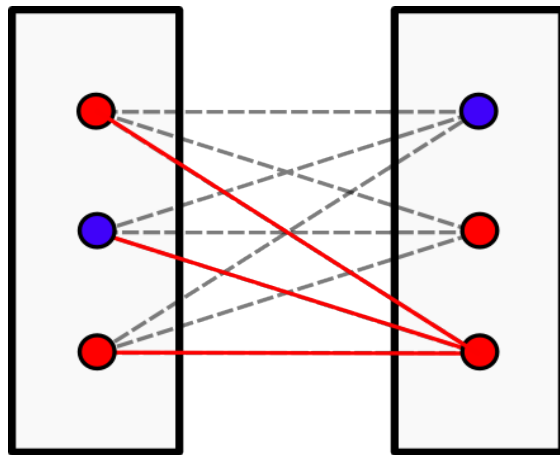
Arc-consistent subset



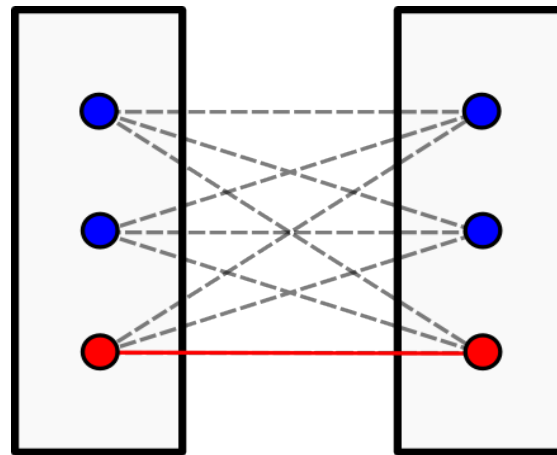
?

- 1) a-c
- 2) not-a-c
- 3) kernel
- 4) no idea

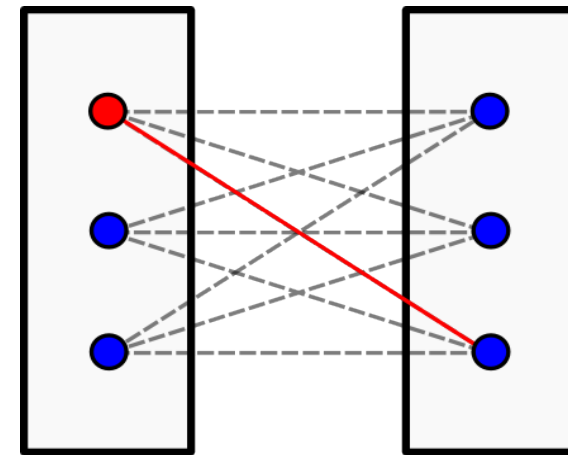
Kernel as the Maximal Arc-Consistent Subset



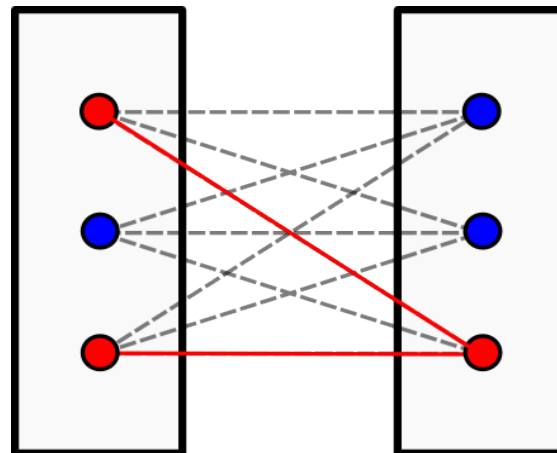
inconsistent



Arc-consistent subset

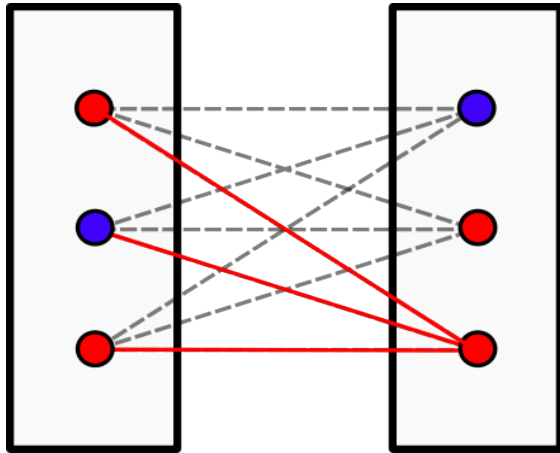


Arc-consistent subset

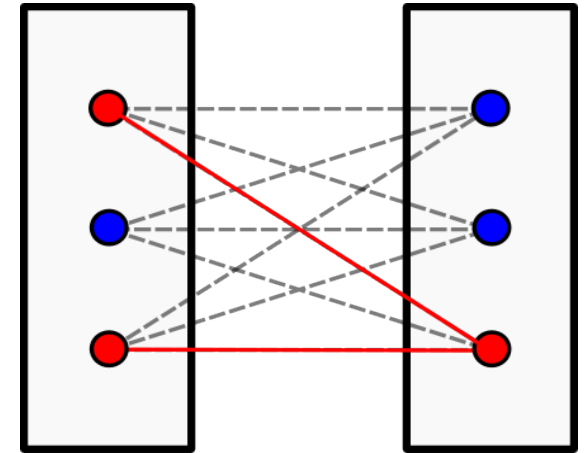


kernel

Kernel as the Maximal Arc-Consistent Subset



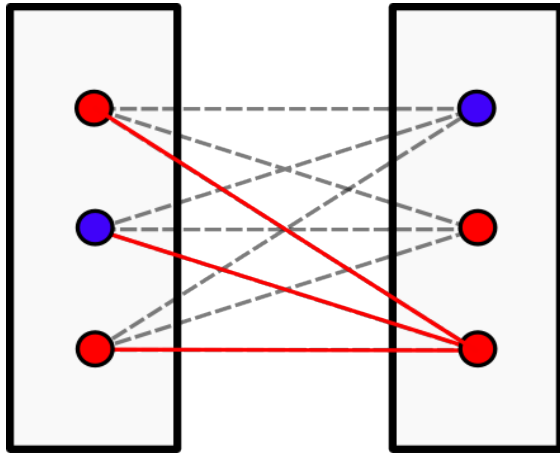
inconsistent



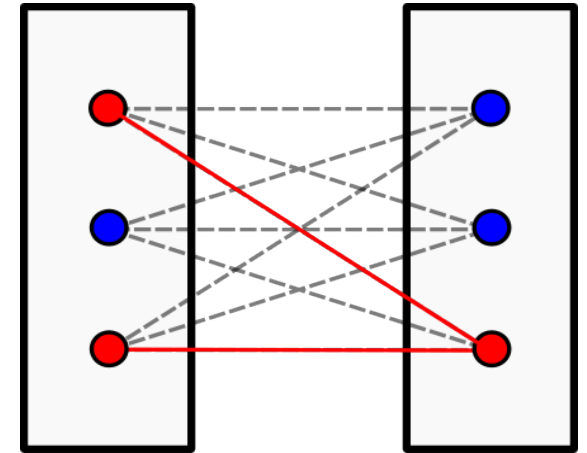
kernel

Can we efficiently find the kernel?

Relaxation Labeling Algorithm



inconsistent



kernel

Can we efficiently find the kernel?

YES – (greedy) Relaxation labeling algorithm

Repeat as long as $\mathbf{z}$ keeps changing:

$$\forall u \in \mathcal{V}, s \in \mathcal{X}_u, v \in \mathcal{N}_b(u): \quad z_u(s) := z_u(s) \wedge \bigvee_{t \in \mathcal{X}_v} z_{uv}(s, t)$$

$$\forall uv \in \mathcal{E}, (s, t) \in \mathcal{X}_{uv}: \quad z_{uv}(s, t) := z_{uv}(s, t) \wedge z_u(s) \wedge z_v(t)$$

Arc-Consistency vs. (Relaxed) Optimality

Any $\delta(x)$, $x \in \mathcal{X}_V$ is arc-consistent

“non-empty kernel of a dual point”



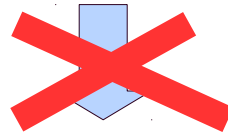
“there is an integer labeling consisting of red edges and labels”

- 1) Yes
- 2) No
- 3) No idea

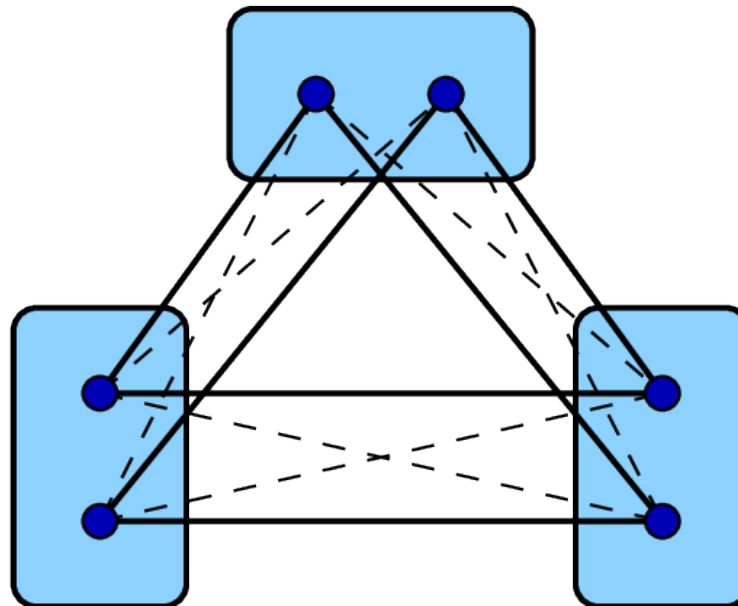
Arc-Consistency vs. (Relaxed) Optimality

Any $\delta(x)$, $x \in \mathcal{X}_v$ is arc-consistent

“non-empty kernel of a dual point”



“there is an integer labeling consisting of red edges and labels”



Any $\delta(x)$, $x \in \mathcal{X}_V$ is arc-consistent

“non-empty kernel of a dual point”

Can we efficiently find out whether:

“there is an integer labeling consisting of red edges and labels”

?

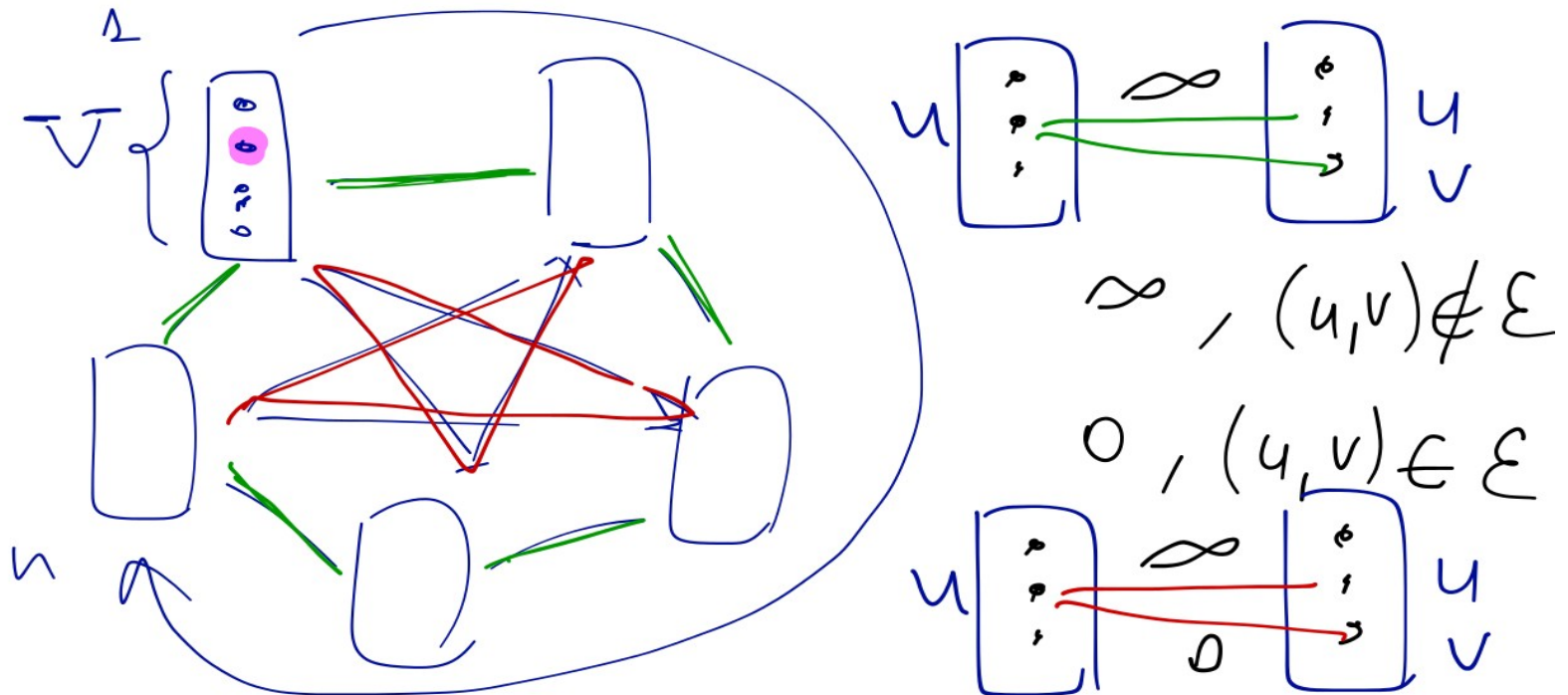
Arc-Consistency vs. (Relaxed) Optimality

Hamiltonian Cycle Reduces to MAP-Inference

Graph $(V, \mathcal{E})$ for a Hamiltonian cycle

Full graph $(V, V \times V)$ for the MAP-inference problem

The set of labels $X_v = V$ for all $v \in V$.



In general – NO. It is an NP-hard Constraint Satisfaction Problem

Arc-Consistency vs. Relaxed Optimality

Any $ce[\mu]$, $\mu \in \mathcal{L}$ is arc-consistent

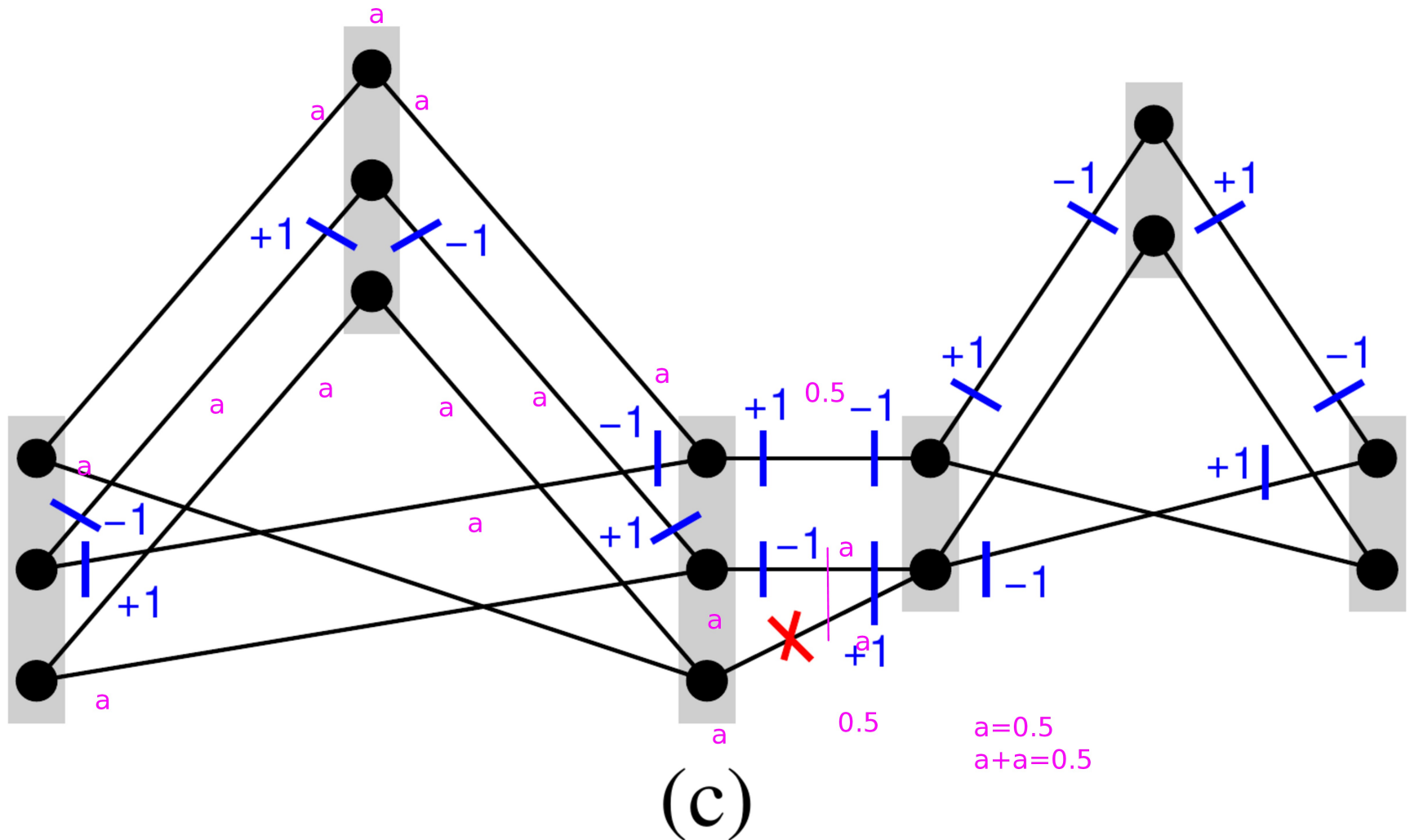
“non-empty kernel of a dual point”



“there is a **relaxed** labeling consisting of red edges and labels”

- 1) Yes
- 2) No
- 3) No idea

Arc-Consistency vs. Relaxed Optimality



Picture: Tomas Werner. A Linear Programming Approach to Max-sum Problem: A Review.

Any $ce[\mu]$, $\mu \in \mathcal{L}$ is arc-consistent

“non-empty kernel of a dual point”

Can we efficiently find out whether:

“there is a **relaxed** labeling consisting of red edges and labels”

?

Arc-Consistency vs. Relaxed Optimality

Any $\text{ce}[\mu]$, $\mu \in \mathcal{L}$ is arc-consistent

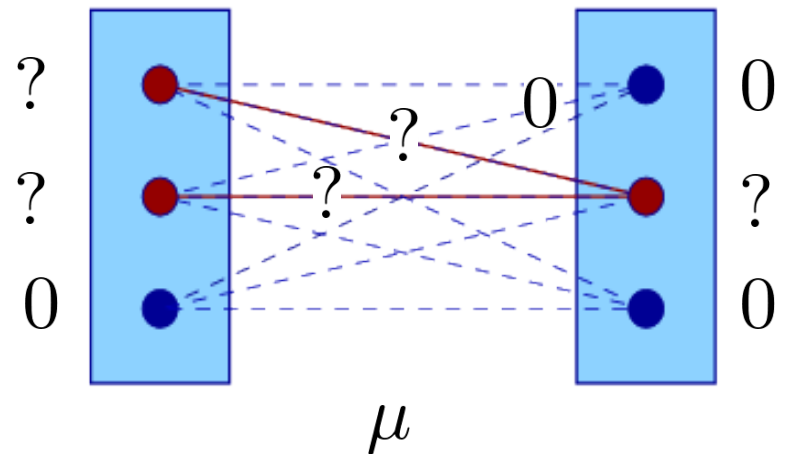
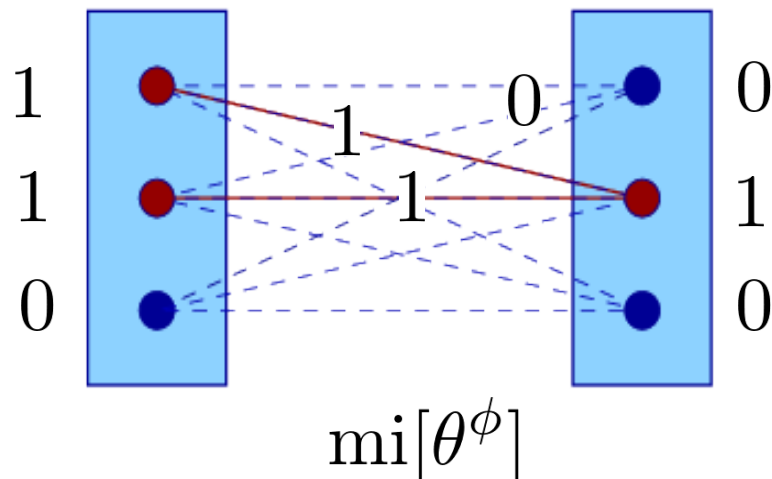
“non-empty kernel of a dual point”

Can we efficiently find out whether:

“there is a **relaxed** labeling consisting of red edges and labels”

?

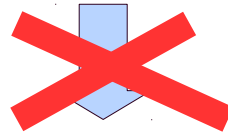
Yes – by solving the reduced LP:



Arc-Consistency vs. Relaxed Optimality

Any $ce[\mu]$, $\mu \in \mathcal{L}$ is arc-consistent

“non-empty kernel of a dual point”



“there is a relaxed labeling consisting of red edges and labels”

Necessary, but not sufficient optimality condition:

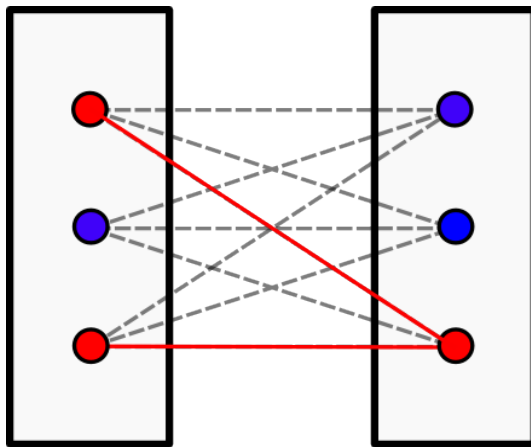
“non-empty kernel of a dual point”

- 1) No optimality**
- 2) No non-strict arc-consistency**

Practice: Dual Rounding

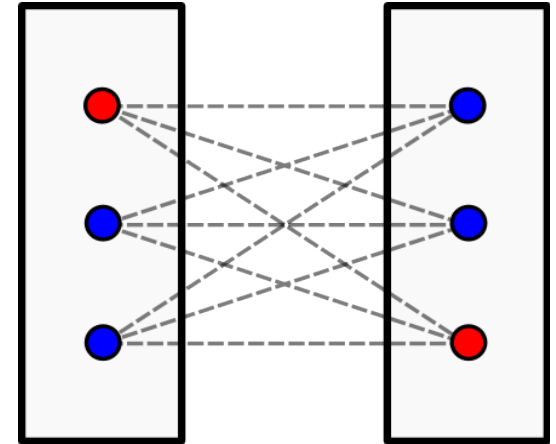
- 1) No optimality
- 2) No non-strict arc-consistency

Kernel of the dual:

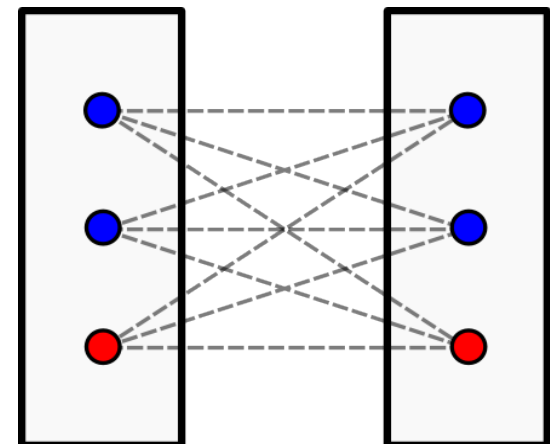


$$\text{mi}[\theta^\phi] = \mathbf{min}$$

Labeling:



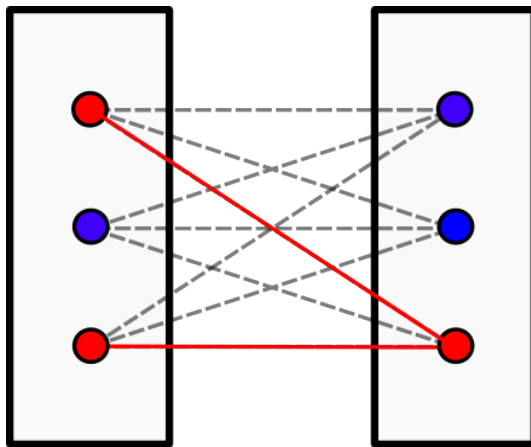
or:



Practice: Dual Rounding

- 1) No optimality
- 2) No non-strict arc-consistency

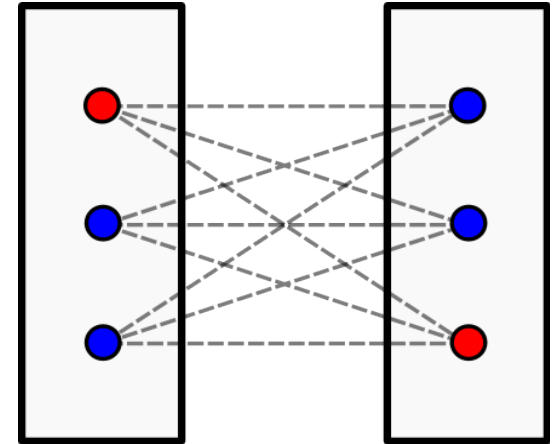
Kernel of the dual:



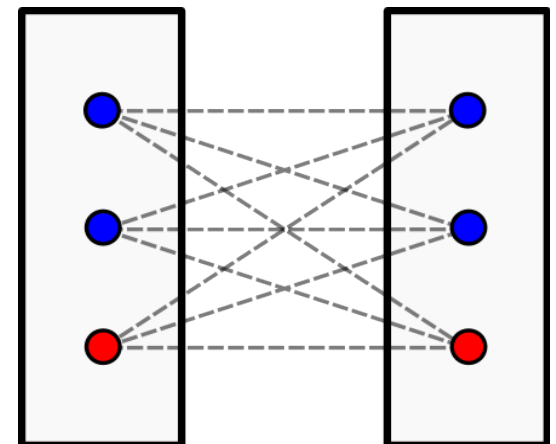
$$\text{mi}[\theta^\phi] = \mathbf{min}$$

No optimality guarantees!

Labeling:



or:



Practice: Dual Rounding



Input Stereopair



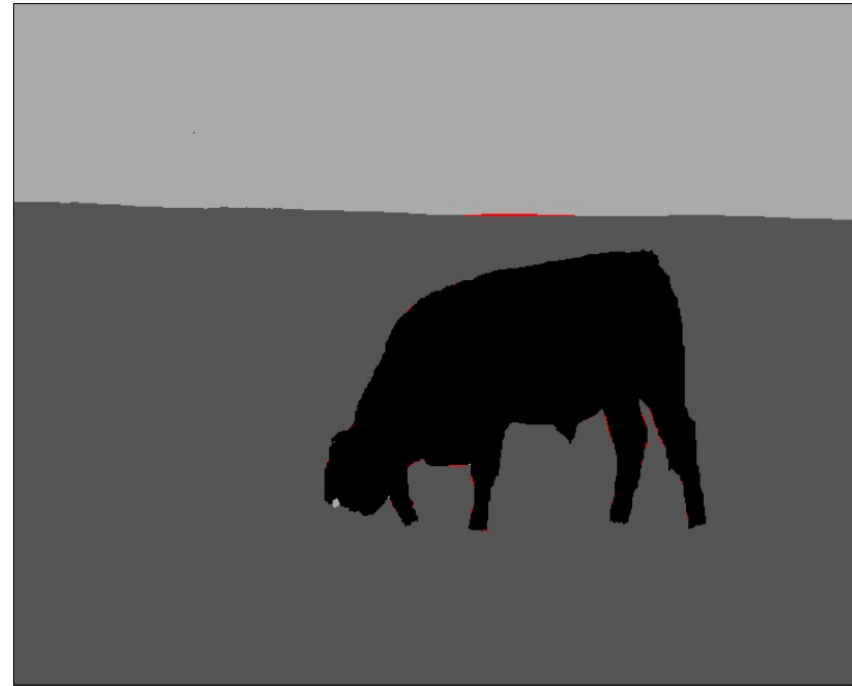
Disparities

Pictures and data: Middlebury Benchmark <http://vision.middlebury.edu>

Practice: Dual Rounding



Input Image



Segmentation

Pictures and data: OpenGM Benchmark <http://hciweb2.iwr.uni-heidelberg.de/opengm>

Practice: Dual Rounding



Multiple images



Stitched image

Pictures and data: Middlebury Benchmark <http://vision.middlebury.edu>

Practice: Dual Rounding

Well, practically good roundings are a bit more tricky ... Stay tuned :).



Multiple images



Stitched image

Pictures and data: Middlebury Benchmark <http://vision.middlebury.edu>